SPHOORTHY ENGINEERING COLLEGE

An Autonomous Institution Approved by AICTE & Accredited with NAAC Grade 'A'
Department of Examination & Academic Affairs · Hyderabad — 501510

Machine Learning: Supervised, Unsupervised & Ensemble Algorithms

Program / Dept: B.Tech / MBA — CSE AIML	**Course Code / Title:** Machine Learning (AI501PC)
Academic Regulation: Autonomous (SR23 / SR24)	**Curriculum Semester:** Semester 5 (Year 3)
Credits / Contact Hours: 3 Credits · 48 Lecture Hours	**Evaluation Scheme:** Continuous Internal (40M) + End Sem (60M)

Course Overview & Pedagogical Objectives:

Linear and Logistic Regression, Gradient Descent optimization, Decision Trees, Random Forests, Support Vector Machines (SVM), K-Means clustering, and Evaluation Metrics. The primary objective of this course is to equip students with sound theoretical rigor, analytical problem-solving competencies, and real-world system engineering capabilities demanded by premier technological organizations and competitive graduate benchmarks.

Course Outcomes (COs) & Bloom's Taxonomy Mapping:

CO ID	Course Learning Outcome Description	Bloom's Level
CO 1	Understand and formulate fundamental concepts of Unit 1: Supervised Regression.	L2 Understand
CO 2	Analyze algorithms, mechanisms, and models in Unit 2: Classification Models.	L4 Analyze
CO 3	Design and evaluate robust architectures for Unit 3: Support Vector Machines & Kernels.	L5 Evaluate
CO 4	Implement and optimize operational pipelines in Unit 4: Ensemble Learning.	L3 Apply
CO 5	Integrate state-of-the-art frameworks in Unit 5: Unsupervised Learning for industrial solutions.	L6 Create

Comprehensive Module Index:

Unit	Title & Core Concepts	Page
Unit 1	**Unit 1: Supervised Regression**: Simple & Multiple Linear Regression, Cost functions, Gradient Descent (Batch, Mini-batch, SGD), Polynomial Reg...	Page 2
Unit 2	**Unit 2: Classification Models**: Logistic regression, Decision Boundary, ROC-AUC curve, Confusion Matrix, Precision-Recall tradeoff, Naive Baye...	Page 3
Unit 3	**Unit 3: Support Vector Machines & Kernels**: Optimal separating hyperplanes, Hard vs Soft margin SVM, Slack variables, Kernel trick (Polynomial, Radial Bas...	Page 4
Unit 4	**Unit 4: Ensemble Learning**: Bagging (Random Forests), Boosting (AdaBoost, Gradient Boosting, XGBoost), Voting Classifiers, Stacking, and B...	Page 5
Unit 5	**Unit 5: Unsupervised Learning**: K-Means clustering algorithm, Elbow method, Hierarchical clustering (Dendrograms), DBSCAN density clustering, ...	Page 6

UNIT 1: SUPERVISED REGRESSION • *Core Theory & Practice Guide*

1.1 Technical Exposition & Conceptual Foundations

The study of unit 1: supervised regression forms an indispensable cornerstone of Machine Learning (AI501PC). Specifically, Simple & Multiple Linear Regression, Cost functions, Gradient Descent (Batch, Mini-batch, SGD), Polynomial Regression, Regularization (Ridge L2, Lasso L1). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

1.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 1 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

1.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 1
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

1.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 1 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 1. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 1 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 1 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 2: CLASSIFICATION MODELS • *Core Theory & Practice Guide*

2.1 Technical Exposition & Conceptual Foundations

The study of unit 2: classification models forms an indispensable cornerstone of Machine Learning (AI501PC). Specifically, Logistic regression, Decision Boundary, ROC-AUC curve, Confusion Matrix, Precision-Recall tradeoff, Naive Bayes classifier, and K-Nearest Neighbors (KNN). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

2.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 2 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

2.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 2
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

2.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 2 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 2. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 2 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 2 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 3: SUPPORT VECTOR MACHINES & KERNELS • *Core Theory & Practice Guide*

3.1 Technical Exposition & Conceptual Foundations

The study of unit 3: support vector machines & kernels forms an indispensable cornerstone of Machine Learning (AI501PC). Specifically, Optimal separating hyperplanes, Hard vs Soft margin SVM, Slack variables, Kernel trick (Polynomial, Radial Basis Function RBF, Sigmoid). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

3.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 3 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

3.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 3
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

3.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 3 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 3. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 3 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 3 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 4: ENSEMBLE LEARNING • *Core Theory & Practice Guide*

4.1 Technical Exposition & Conceptual Foundations

The study of unit 4: ensemble learning forms an indispensable cornerstone of Machine Learning (AI501PC). Specifically, Bagging (Random Forests), Boosting (AdaBoost, Gradient Boosting, XGBoost), Voting Classifiers, Stacking, and Bias-Variance tradeoff analysis. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

4.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 4 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

4.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 4
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

4.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 4 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 4. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 4 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 4 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 5: UNSUPERVISED LEARNING • *Core Theory & Practice Guide*

5.1 Technical Exposition & Conceptual Foundations

The study of unit 5: unsupervised learning forms an indispensable cornerstone of Machine Learning (AI501PC). Specifically, K-Means clustering algorithm, Elbow method, Hierarchical clustering (Dendrograms), DBSCAN density clustering, and Principal Component Analysis (PCA). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

5.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 5 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

5.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 5
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

5.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 5 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 5. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 5 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 5 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

Standard Recommended Textbooks & Digital References:

1. *Primary Curriculum Standard Textbook* — Pearson Higher Education / McGraw-Hill Education (Latest Edition).
2. *Autonomous Reference Syllabus Blueprint* — Sphoorthy Engineering College Academic Council.

3. *NPTEL & IEEE Computer Society Guidelines* — Open Educational Resources & Peer-reviewed Publications.